

Skills Summary

One-Way Implications Between Quadrilaterals

Use this reference while completing your worksheet or when studying.

The family tree — read the arrows in one direction only:

square $\Rightarrow$ rectangle $\Rightarrow$ parallelogram $\Rightarrow$ trapezoid and square $\Rightarrow$ rhombus $\Rightarrow$ kite

Every square is a rhombus. **Not every** rhombus is a square. Going *down* the tree is free; going *up* always costs one extra condition, and you must say which one you have.

Skill 1 — Angle tests in a parallelogram

What a parallelogram guarantees: opposite angles congruent, and consecutive angles supplementary (they add to 180°).

What it does not guarantee: right angles. That is the extra condition that upgrades it to a rectangle.

Example: In a parallelogram, $m\angle A = (4x + 10)^\circ$ and $m\angle B = (2x + 50)^\circ$ are consecutive. Find x .

Step 1. Consecutive angles of a parallelogram are supplementary — so add the two expressions and set the total to a straight angle, rather than setting either one to a right angle.

Step 2. $(4x + 10) + (2x + 50) = 180$, so $6x + 60 = 180$ and $6x = 120$
 $\Rightarrow x = 20$

Try it: Consecutive angles measure $(3x + 25)^\circ$ and $(x + 15)^\circ$. Find x .

check: 38

Skill 2 — Side-length tests

Rules: all four sides congruent $\Rightarrow$ rhombus. Two pairs of *adjacent* sides congruent $\Rightarrow$ kite. One pair of opposite sides both parallel and congruent $\Rightarrow$ parallelogram.

Congruent sides never produce right angles, so no side test alone can reach rectangle or square.

Example: A rhombus has $PQ = 5x - 2$ and $QR = 3x + 8$. Find x .

Step 1. All sides of a rhombus are congruent, so the two expressions describe the same length — set them equal instead of adding them.

Step 2. $5x - 2 = 3x + 8$, so $2x = 10$
 $\Rightarrow x = 5$

Try it: A rhombus has $PQ = 7x - 3$ and $QR = 4x + 18$. Find x .

check: 7

Skill 3 — Diagonal tests

Rules: diagonals bisect each other $\Rightarrow$ parallelogram. Add *congruent* diagonals $\Rightarrow$ rectangle. Add *perpendicular* diagonals $\Rightarrow$ rhombus. Both extras together $\Rightarrow$ square.

In a rhombus the diagonals are perpendicular bisectors, so each half-diagonal pair gives a right triangle whose hypotenuse is a side.

Example: A rhombus has half-diagonals 5 and 12 at the centre. Find a side.

Step 1. The diagonals of a rhombus are perpendicular, so a side is the hypotenuse of a right triangle on the two half-diagonals — that is the pythagorean theorem, not an addition.

Step 2. side = $\sqrt{5^2 + 12^2} = \sqrt{169}$
 $\Rightarrow$ **13**

Try it: A rhombus has half-diagonals 8 and 15. Find a side.

check: 17